Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1	*(a)*	(i)		174, 180, 194, 196 – all correct		1		1	1	1
		(ii)		50		1		1	1	1
		(iii)		3 [throws]		1		1	1	1
	(b)			14 (1) 7 (1)		2		2	2	
	(c)	(i)		800 in every cell in the second column (1) 130, 70, 40, 20, 10 in last column (1)		2		2	2	
		(ii)		Plots at (15,130), (20,70), (25,40), (30,20), (35,10) **ecf -** within a tolerance of $\pm$ <1 small square 2 marks for all 5 plots correct 1 mark for 4 plots correct 0 marks for 3 or fewer plots correct Smooth curve must extend back to 800 judge quality of curve in the region 10 – 35 thousand years it must pass within $\pm$ <1 small square of these points (1) Don't accept double lines, whispy, disjointed curves		3		3	2	
	(d)	(i)		It is the time taken to halve (1) the {number of radioactive particles or atoms or nuclei / activity / mass / amount of the substance} (1)	2			2		
		(ii)		6 $\pm$ 0.5 [thousand years]		1		1	1	

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)			[The 60 (6) million fall [out of 800 (80) million] in carbon nuclei or rise in nitrogen nuclei] occurs at about 700 years (1) Which is less than 2 000 years so the claim is incorrect (1) **Alternative:** After 2 000 years approx. 160 / 170 (16 / 17) million [out of 800 (80) million] nuclei of carbon will have decayed / nitrogen nuclei produced (1) So 60 (6) million nuclei will have decayed in much less time (1)			2	2	2	
				Question 1 total	**2**	**11**	**2**	**15**	**12**	**3**